

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2019-P1-Model)** **Date: 01-02-2026**

Time: 09.00Am to 12.00Pm **GTA-13** **Max. Marks: 186**

01-02-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P1)_GTA-13_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2019-P1-Model

Time: 3:00Hr's | IMPORTANT INSTRUCTIONS | Max Marks: 186

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 5 – 12)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 22)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 23 – 30)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 40)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 41 – 48)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

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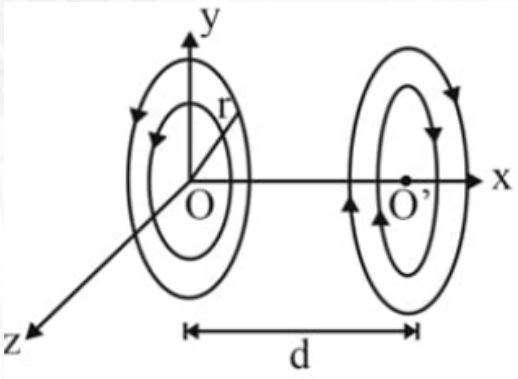
PHYSICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

1. A long uniform cylindrical beam of radius R consists of positively charged particles each of charge q, mass m and velocity V along positive x direction. The axis of the beam is the x-axis. The beam is incident on a region having magnetic field in y-z plane. The magnetic field in the region is confined to $0 \le x \le \Delta x$ and $d \le x \le d + 2\Delta x$, ($\Delta x$ is very small). The magnetic field lines are circular in yz plane as shown. The magnitude of the field is given by $B = B_0 r$ where B_0 is a constant and r is distance from the x-axis. [Assume no interaction among charges within the beam]



A) If $d = \dfrac{mV}{2qB_0\Delta x}$, then final emergent beam is parallel

B) If $d = \dfrac{mV}{2qB_0\Delta x}$, then final emergent beam is convergent

C) If $d = \dfrac{mV}{qB_0\Delta x}$, then final emergent beam is convergent

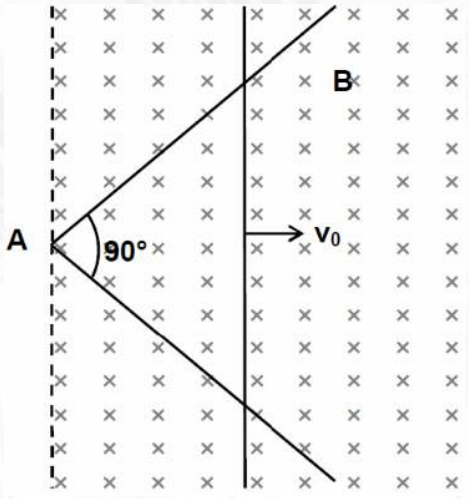
D) None of these

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2. A particle of mass m is moving in the xy-plane such that its velocity at a point (x, y) is given as $\vec{v}=\alpha\left(y\hat{x}+x\hat{y}\right)$ where α is a non-zero constant. What is the torque acting on the particle about the origin ?

A) $\vec{0}$ **B)** $2m\alpha xy\hat{k}$ **C)** $m\alpha xy\hat{k}$ **D)** $-2\alpha xy\hat{k}$

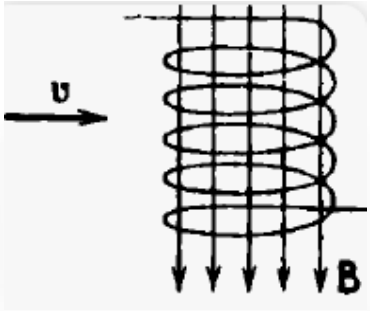
3. A conducting uniform wire bent into the shape of a right angle is held stationary on a horizontal frictionless surface. A very long uniform thin conducting rod of mass m projected with velocity v_0 from the apex A of the bend wire. The resistance per unit length of the wire and the rod is λ. The entire arrangement is placed in a uniform magnetic field B perpendicular to the plane formed by the wire and the rod. Find the distance travelled by the rod before it becomes stationary.



A) $\sqrt{\frac{m\lambda v_0(\sqrt{2}+1)}{B}}$ **B)** $\frac{\sqrt{m\lambda v_0(\sqrt{2}+1)}}{2B}$ **C)** $\frac{\sqrt{m\lambda v_0(\sqrt{2}+1)}}{B}$ **D)** $\sqrt{\frac{m\lambda v_0(\sqrt{2}+1)}{2B}}$

4. A direct current flowing through the winding of a long cylindrical solenoid of radius R produces in it a uniform magnetic field of induction B. An electron files into the solenoid along the radius between its turns (at right angles to the solenoid axis) at a velocity v as shown in figure. After a certain time, the electron deflected by the magnetic field leaves the solenoid. Then the time *t* during which the electron moves in the solenoid is

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A) $\frac{m}{eB}\tan^{-1}\left(\frac{eBR}{mv}\right)$ B) $\frac{2m}{eB}\tan^{-1}\left(\frac{eBR}{mv}\right)$ C) $\frac{m}{eB}\tan^{-1}\left(\frac{mv}{eBR}\right)$ D) $\frac{2m}{eB}\tan^{-1}\left(\frac{mv}{eBR}\right)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

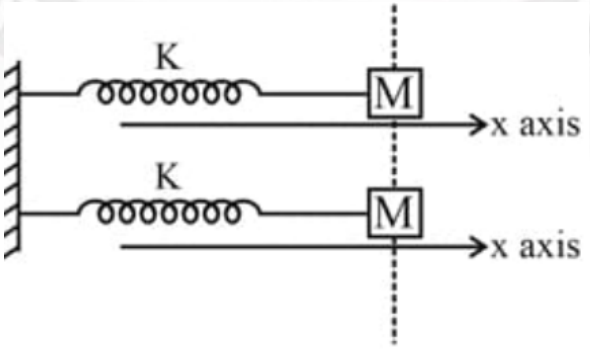
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

5. There are two identical spring block systems, placed on horizontal smooth table. As shown in figure, Now block A is displaced in positive x axis by distance a_0 and B is placed $\frac{a_0}{2\sqrt{3}}$ distance in negative x axis. At t = 0, block A is left free & block B is given velocity $\frac{a_0}{2}\sqrt{\frac{K}{M}}$ in positive x direction, then



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A) Time taken by both the particles to come into same position first time will be

$$\left[\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)\right]\sqrt{\frac{K}{M}}\sec.$$

B) Time taken by particles to decrease distance between them upto minimum possible value will be $\left[\frac{\pi}{2}+\tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)\right]\sqrt{\frac{K}{M}}\sec.$

C) Time taken by particles to have their relative velocity minimum is

$$\left[\frac{\pi}{4}+\tan^{-1}\left(\frac{1}{3\sqrt{3}}\right)\right]\sqrt{\frac{K}{M}}\sec.$$

D) Particles will never come in same phase.

6. A conducting spherical shell of inner radius a and outer radius b is made up of a material of resistivity ρ and negligible dielectric activity. At time t=0, material of shell is electrically netural and a charge q_0 is some how located at the centre of shell (i.e. at time t = 0, there are no induced charges, they will appear afterward). Choose correct statement(s)

A) When $-Q_t$ charge is induced at inner surface at time t, then electric field in the material of shell is $E=\frac{q_0-Q_t}{4\pi\in_0 r^2}$ where r is distance of point under consideration from centre of shell.

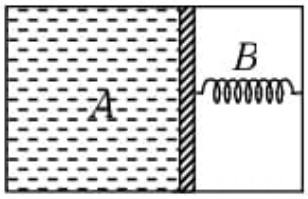
B) Current through shell at this moment (discussed in option A) is $I=\frac{q_0-Q_t}{\in_0\rho}$

C) At time t we can write $\frac{dQ_t}{q_0-Q_t}=\frac{dt}{\in_0\rho}$

D) At time t, value of Q_t is given by $Q_t=q_0\left(1-e^{-\left(t/\in_0\rho\right)}\right)$

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7. A thermally insulated chamber of volume $2V_0$ is divided by a frictionless piston of area S into two equal parts A and B. Part A has an ideal gas at pressure P_0 and temperature T_0 and in part B is vacuum. A massless spring of force constant k is connected with piston and the wall of the container as shown. Initially spring is unstretched. Gas in chamber A is allowed to expand. Let in equilibrium spring is compressed by x_0. Then



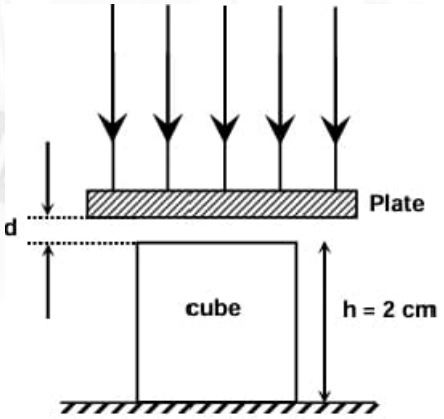
A) final pressure of the gas is $\frac{kx_0}{S}$

B) work done by the gas is $\frac{1}{2}kx_0^2$

C) change in internal energy of the gas is $-\frac{1}{2}kx_0^2$

D) temperature of the gas is decreased.

8. An electromagnetic radiation of wavelength ranging between 400 nm and 1150 nm (for which the plate is penetrable) incident for perpendicularly on the plate from above, is reflected from both the air surfaces and interferes. In this range only two wavelength give maxima, one of them is $\lambda = 400\,nm.$ (refractive index of air = 1 and the coefficient of linear thermal expansion of cube $= 8\times10^{-6}\ {}^{o}C^{-1}$). The distance of bottom of cube from plate does not change during warming up. Choose the correct options(s)



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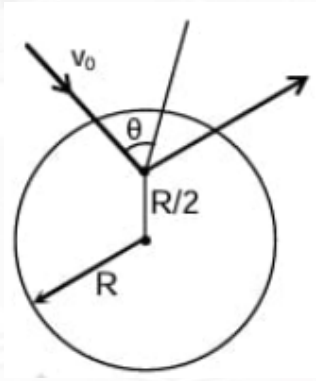
A) The change in temperature of cube so that it would touch the plate is $6.2^{o}C$

B) The change in temperature of cube so that it would touch the plate is $3.1^{o}C$

C) The second wavelength, which gives maxima is $\lambda = \frac{2000}{3} nm$

D) The second wavelength, which gives maxima is $\lambda = \frac{4000}{3} nm$

9. A disc of radius R = 20 cm and mass m is lying (in X-Z plane) on smooth horizontal surface. A particle of mass m moving with velocity $\vec{v}_0 = \left(5\hat{i} - 10\hat{j}\right) m/s$ collides with the disc at a distance $\frac{R}{2}$ from centre of the disc. The coefficient of restitution e = 1 and coefficient of friction between particle and disc is $\mu = 0.2$. Assume slipping during collision and the vertical plane in which particle is moving is at distance R/2 from centre of disc. The correct statements are $(g = 10 m/s^2)$



A) Angular velocity of disc after collision is 20 rad/s

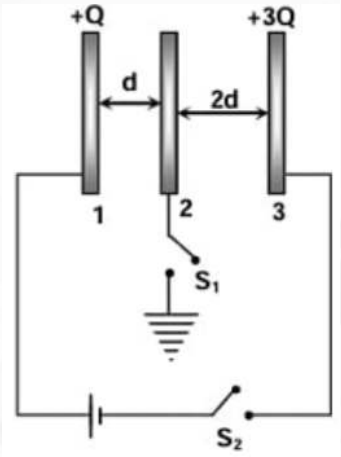
B) Linear velocity of the centre of disc is 4 m/s

C) The ratio of time of flight of particle after collision and time period of disc is $\frac{20}{\pi}$

D) The ratio of time of flight of particle after collision to time period of disc is $\frac{\pi}{20}$

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10. Three conducting sheets 1, 2 and 3 are placed parallel to each other such that separation between them is negligible. Sheet-1 and sheet-3 are given charges +Q and +3Q respectively. Cell of emf 1 V is to be connected between sheet 1 and 3 (Take $\frac{\epsilon_0 A}{d} = 30\ mF$ and Q = 3 mC) where A is area of sheets. First switch S_1 is closed and then switch S_2 is closed.



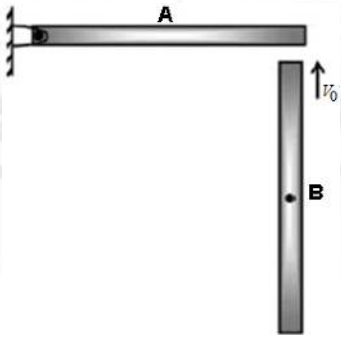
A) When switch S_1 is closed charge flow through S_1 is zero.

B) When switch S_1 is closed charge flow through S_1 is 12 mC.

C) When S_2 is closed charge flown through S_2 is 2 mC.

D) When S_2 is closed charge flown through S_2 is 15 mC.

11. In figure shown rod A of mass 2m and length $2l$ is kept on a horizontal frictional surface and hinged at point O such that it can rotate about vertical axis passing through O. Another identical rod B moving with a constant velocity v_0 collides with rod A at its end and stickes to it (both rods are perpendicular to each other)



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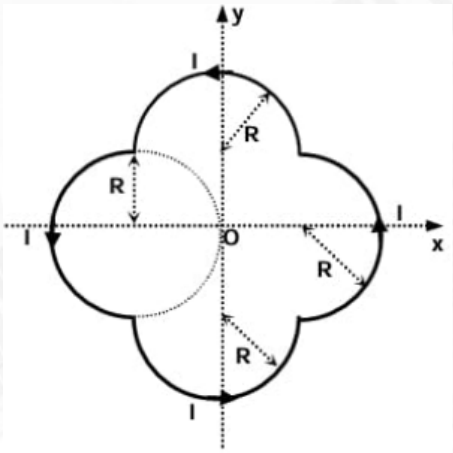
A) angular velocity of system just after collision is $\frac{3v_0}{10l}$

B) Velocity of centre of mass at system just after collision is $9v_0\sqrt{\frac{2}{5}}$

C) Centre of mass of system is at a distance of $\sqrt{\frac{5}{2}}l$ from O.

D) Kinetic energy of system just after collision is $\frac{3mv_0^2}{5}$

12. A system of four semi-circular wires of radius a, carrying current I as shown in the figure



A) The magnetic field at 'O' due to current carrying loop is $\frac{\mu_0 I}{2\pi R}l\,n(\sqrt{2}+1)(\hat{k})$

B) The magnetic field at 'O' due to current carrying loop is $\frac{\mu_0 I}{\pi R}l\,n(\sqrt{2}+1)(\hat{k})$

C) The magnetic moment of the system is $4R^2I(2+\pi)(+\hat{k})$

D) The magnetic moment of the system is $2R^2I(2+\pi)(+\hat{k})$

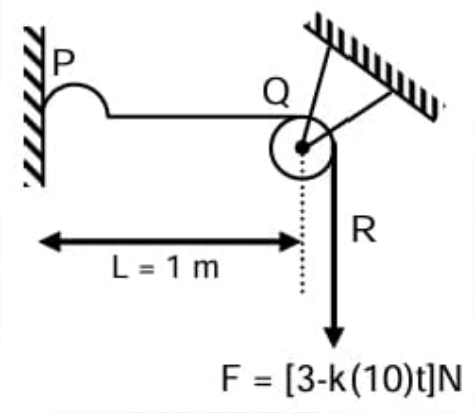
SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

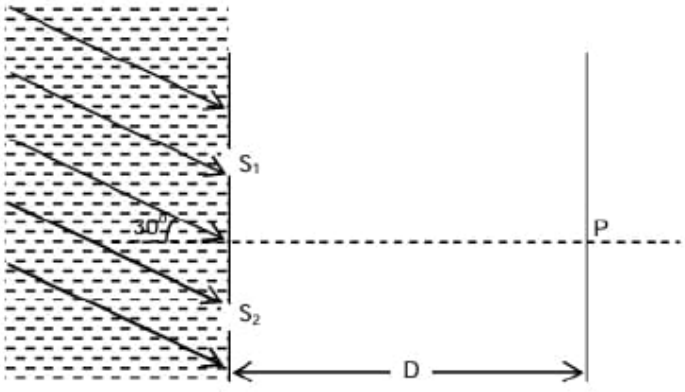
- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

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13. As shown in figure light string PQR is stretched by force F = (3 – 10kt) N, where k is a constant and t is time in second. At time t = 0, a pulse is generated at the end P of the string. For the value of k (in N/s) if the value of force becomes zero as the pulse reaches point Q. Given mass unit length of string is 0.03 kg/m.

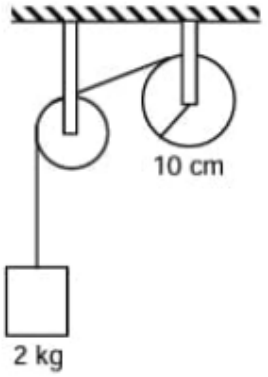


14. The given figure shows a YDSE apparatus. Parallel monochromatic coherent light rays are incident on slits S_1 *and* $S_2 (S_1S_2 = \frac{2}{3} mm)$ at an angle 30° with the horizontal. The medium on left side of the slits is water $(\mu_w = 4/3)$. To obtain the central maxima at point P a glass slab $(\mu_g = 3/2)$ inside water is introduced in front of slit S_1. The thickness of the glass slab required for this purpose (in mm) is $\frac{x}{3}$, then find the value of 'x'.



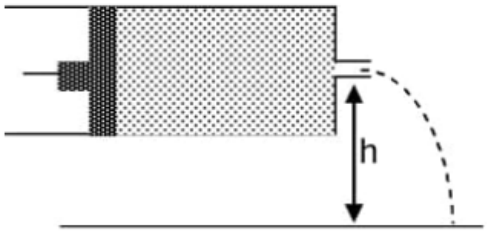
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15. A string is wrapped on a wheel of moment of inertia $0.20\,kg\,m^2$ and radius 10 cm and goes through a light pulley to support a block of mass 2.0 kg as shown in figure. If $g = 10 m/s^2$, then the acceleration of the block $(in\, m/s^2)$ is $\frac{2x}{11}$, then calculate the value of 'x'.



16. A bob of mass m, suspended by a string of length l_1, is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it colloids elastically with another bob of mass m suspended by a string of length l_2, which is initially at rest. Both the strings are mass-less and inextensible. If the second bob, after collision acquires the minimum speed required to complete a full circle in the vertical plane, the ratio $\frac{l_1}{l_2}$ is

17. A tube has two area of cross-sections as shown in figure. The diameter of the tube are 8mm and 2mm. Find range (in m) of water falling on horizontal surface, if piston is moving with a constant velocity of 0.25 m/s, h = 1.25m $(g = 10 m/s^2)$



18. Two rods are of same material and have same length and area. Heat ΔQ flow through them in 12 minutes, when they joined side by side. If now both the rods are joined in parallel, then in what time (in minutes) the same amount of heat ΔQ will flow.

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CHEMISTRY **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

19. In which of the following cationic analysis the mentioned reagent does not undergo redox change?

A) $Cu^{2+}, aq\,KI$ **B)** $Co^{2+}, aq\,KNO_2$ **C)** $Fe^{3+}/H^+, H_2S$ **D)** Fe^{2+}/KCN

20. In which metallurgical process, formation of a complex of the metal ion is not involved ?

A) Leaching of Al_2O_3, Fe_2O_3 in alkali

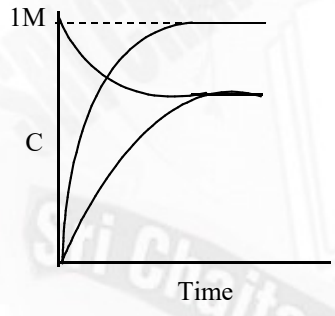
B) Seperation of ZnS and PbS in froth floatation with NaCN depressant

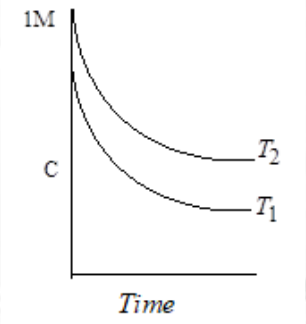
C) Refining of Nickel in Mond's process.

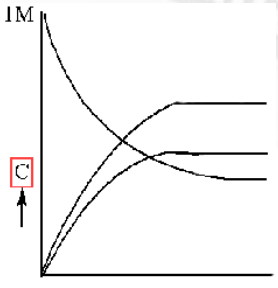
D) Electrolytic reduction of pure alumina in cryolite

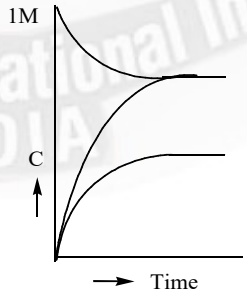
21. $A_2B_3(g) \rightleftharpoons A_2B(g) + 2B(g)\ \ \Delta H > 0$

If $\Delta G^o > 0$ and $T_2 > T_1$ which of the following graphs may be correct ? (closed rigid container started with fresh A_2B_3)

A)


B)


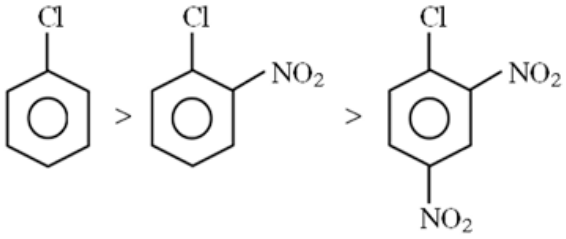
C)


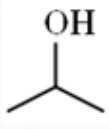
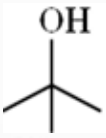
D)


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22. Which of the following is/are CORRECT ?

A) Rate of ArS_N2



B) Compounds  and  can not be differentiated using iodoform test

C) In case of Pyridine negative end of dipole on nitrogen and positive end towards nucleus of ring where as in case of pyrrole the positive end of dipole on nitrogen and the negative end towards nucleus of ring.

D) Nitration of Halobenzenes with concentrated $HNO_3 + H_2SO_4$, among Halobenzenes, PhI gives more percentage of para product

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

23. Which comparisons are correct ?

A) $\left[Co(en)_3\right]^{3-} > \left[Co(NH_3)_6\right]^{3+}$:$CFSE$

B) $\left[Rh(NH_3)_6\right]^{3+} > \left[Ir(NH_3)_6\right]^{3+}$: Wave length of absoption spectral line.

C) $\left[V(CO)_6\right]^{-} > \left[Cr(CO)_6\right]$: 'C-O' bond length

D) $\left[Pt(NO_2)_2(NH_3)_2\right] > \left[Zn(NO_2)_2(NH_3)_2\right]$: Number of possible isomers.

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24. In which electrolytes the oxygen is oxidized during electrolysis ?

A) $50\%\, H_2SO_4\, aq$ **B)** $aq\, KI$

C) $CH_3COOK(aq)$ **D)** $aq.K_2SO_4$

25. In which reaction (s) a new (p-d) π bond is developed during formation of products ?

A) hydrolysis of SCl_4 **B)** disproportionation of ortho phosphorous acid

C) Catalytic oxygenation of SO_2 **D)** Passing Cl_2 gas into hot concentrated KOH

26. Which of the following statement(s) is (are) correct assuming ideal gas behavior ?

A) The r.m.s velocity of 'He' can be doubled by increasing the pressure four fold at 300 K.

B) In a rigid closed container the mean free path of a molecule is doubled by making the absolute temperature doubled.

C) 'SO_2' and 'He' posses same average translational kinetic energy at a given temperature.

D) For a gas with symmetric velocity distribution curve both most probable velocity and average velocity are equal for a given gas.

27. In which molecules all bond lengths are not equal

A) PCl_5 **B)** $CrCl_6^{4-}$ **C)** $CuCl_6^{2-}$ **D)** $Cr_2O_7^{2-}$

28. Radio active samples follow 1st order kinetics during their disintegration. Which statements are correct in view of the salts of Radioactive metals ? (Ra and U are radio active)

A) One gram Radium chloride has greater radio activity than one gram Radium oxide

B) One mole Radium nitride has greater radioactivity than one mole Radium oxide.

C) A rock originally uranium, now containing "Pb / U" mole ratio as "3" is aged about two half lives of 'U'

D) Saturated solutions of Radium phosphate $\left(K_{sp} = x\right)$ and Radium sulphate $\left(K_{sp} = 3x\right)$ show same rate of decay.

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29. Total number of benzene ring containing racemic mixtures possible with molecular formula $C_8H_{10}O_2$ which can also gives violet colour with neutral $FeCl_3$

A) 3 **B)** 6 **C)** 9 4

30. $(P) \xrightarrow{Conc.HBr} (P_1) \xrightarrow{Na/Dry\ ether} (P_2) \xrightarrow{Cl_2/h\upsilon} (P_3) \xleftarrow{SOCl_2} (Q)$

$$\underset{\text{(optically active alcohol)}}{\overset{Q}{}} \xrightarrow[ii)CH_3-Br]{i)\ Na} \text{2- Methoxy Butane}$$

Choose the correct statements :

A) P is a 2° alcohol

B) P_2 is heated with anhydrous $AlCl_3, HCl$ to give isobutane.

C) Reaction of Q with $SOCl_2$ involve SN^1 Mechanism

D) P & Q will take same time to make turbidity with Lucas reagent.

SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

31. Study the following outline reaction sequence (unbalanced).

$$P_4 \xrightarrow[\Delta]{aq.NaOH} \underset{(aq)}{A} + \underset{(g)}{B}\uparrow$$

$$\underset{(aq)}{A} \xrightarrow{HCl} C + NaCl$$

$$\underset{aq}{C} \xrightarrow{\Delta} \underset{(tribasic)}{D} + B\uparrow$$

If the sequence is started with 0.1 moles of P_4, how much volume of 0.1 M NaOH in litres is required to completely neutralize the quantity of 'D' formed ? (All the reactions yield 100% products)

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32. XeF_4 undergoes hydrolysis as per the given balanced reaction.

$$6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$$

To account for the electron flow in the reaction, in the total XeF_4 content $x\%$ is only undergoing disproportionation and the remaining is reduced by water. What is 'x' ?

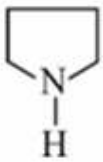
33. E^o_{cell} Value of $A^+, A^{2+} // A^+ / A$ cell is 0.5 volts. What is the value of E^0_{cell} *of* $A / A^{2+} // A^+ / A$ in volts at same temperature ?

34. $AX_2 \rightarrow A^{+2} + 2X^- \quad \Delta H = +4\,Kcal\,mol^{-1}$

If 1 mole of fresh sample of AX_2 is allowed to ionize in five kilograms of water in α calorimeter (heat capacity of the contents together $= 300 cal\,K^{-1}$). If the dissociation follows 1st order kinetics, after 16 minutes from the beginning the temperature of contents is dropped by $10^o C$. What could be the theoretical abnormal boiling point $\left({}^oC\right)$ of the medium at the end of 8th minute from the beginning ?

(ebuloiscopic constant of water $= 0.50\,K\,kg\,mol^{-1}$) ($P_{atm} = 1$)

35. 2.5 litres of an H_2O_2 solution is treated with an acidified $KMnO_4$ solution until 60% of H_2O_2 is oxidized. The remaining 'H_2O_2' in the solution is thermally decomposed. In the entire process 2.24 lts of O_2 is collected at STP. How many moles of '$KMnO_4$' is consumed in the process ?

36. 

$$\xrightarrow[(3)\ \Delta]{\substack{(1)\ CH_3I(excess)\\(2)\ Moist\ Ag_2O}} \xrightarrow[(3)\ \Delta]{\substack{(1)\ CH_3I(excess)\\(2)\ Moist\ Ag_2O}} P + Nitrogen\,containing\,compound$$

A) Degree of unsaturation of product 'P' is 'X'.

B) Number of different types of carbon - carbon bond lengths in 'P' is 'Y'

C) Over all total number of moles of CH_3I consumed is 'Z'.

Then calculate the value of product of xyz is

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MATHEMATICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

37. If $f(x)=\left(p^2-1\right)\left[\tan^{-1}x\right]+4\left(q^2+2q-3\right)\left\{\frac{1}{2+x^2}\right\}+(p+q)\operatorname{sgn}\left(x^2-x+2\right)$ is continuous in R and $f(x_1)=f(x_2)\forall x_1,x_2\in R$ then largest value of $|p+q|$ is

[Note: sgn(y), [y] and {y} denote signum function, greatest integer function and fractional part function respectively.]

A) 0 **B)** 2 **C)** 4 **D)** 5

38. ${}^{101}C_0.{}^{100}C_{100}+{}^{101}C_2.{}^{100}C_{98}+{}^{101}C_4{}^{100}C_{96}+.....{}^{101}C_{100}{}^{100}C_0=$

A) ${}^{201}C_{100}$ **B)** ${}^{201}C_{100}+{}^{100}C_{50}$

C) $\frac{{}^{201}C_{100}-{}^{100}C_{50}}{2}$ **D)** $\frac{{}^{201}C_{100}+{}^{100}C_{50}}{2}$

39. Consider a point O (6, 2) on a circle $x^2+y^2-4x-10y+4=0$. Two lines $x-7y+8=0$ and $x-2y-2=0$ are drawn from point O, which intersect the given circle at A and B respectively. If slope of line AB is M, then |M| =

A) $\frac{1}{3}$ **B)** $\frac{1}{4}$ **C)** $\frac{2}{3}$ **D)** 1

40. If $8\int_0^{\pi/2}\frac{x^2(\cos x-\sin x)dx}{(\cos x+\sin x)^3}=\pi(k-\pi)$, then k is equal to

A) 3 **B)** 2 **C)** 0 **D)** 5

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

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41. Let $I=\int_1^2 \frac{e^x+e^{4/x}}{x}dx,$ then which of the following is/are correct statement(s) ?

A) $2I=\int_1^4 \frac{e^x+e^{4/x}}{x}dx$

B) $4I=\int_1^4 \frac{e^x+e^{4/x}}{x}dx$

C) $I>\left(e^2-e\right)$

D) $I<\left(e^2-e\right)$

42. Let $L_1: x+y=1, z+2y=2$ and $L_2: x-y=1, z-2y=2$ be two straight lines, then which of the following is/are correct statements(s) ?

A) the line L_1 and L_2 are coplanar

B) the lines L_1 and L_2 are non-coplanar

C) on the line $\vec{r}=4\hat{j}+\lambda\left(\hat{i}+2\hat{k}\right), \lambda\in R,$ there exist infinite number of points such that an infinite number of lines passing through each of these points and intersecting both the lines L_1 and L_2 can be drawn

D) there exists a unique line passing through the point $A(1,\mu,3)$ which intersect both lines L_1 and $L_2 \forall \mu\in R$

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43. Let $P=\sum_{n=1}^{10} n\left\{\frac{1^2}{1+n}+\frac{2^2}{2+n}+.....+\frac{10^2}{10+n}\right\}$ then which of the following is/are correct ?

A) 2P is perfect square

B) 2P is divisible by 11^2

C) Number of positive integral divisors of 2P is 9

D) The remainder when 2P is divided by 6 is 1

44. Suppose f(x) be a twice differentiable polynomial function such that f(1) = 2, f(2) = 8 and f(3) = 18, then

A) $f''(x)=4\forall x\in R$

B) there exist at least one $x\in(1,3)$ such that $f''(x)=4$

C) there exist at least one $x\in(2,3)$ such that $f'(x)-10=0$

D) there exist at least one $x\in(1,3)$ such that $f'(x)-8=0$

45. Which of the following alternative(s) is/are incorrect ?

A) Number of values of x satisfying simultaneously $\sin^{-1}x=2\tan^{-1}x$ and

$\tan^{-1}\sqrt{x(x-1)}+\cos ec^{-1}\sqrt{1+x-x^2}=\frac{\pi}{2}$, *is* 2

B) Number of values of x satisfying simultaneously

$\sin^{-1}x=2\tan^{-1}x$ and $\tan^{-1}\sqrt{x(x-1)}+\cos ec^{-1}\sqrt{1+x-x^2}=\frac{\pi}{2}$, is greater than 2

C) The complete solution set of the equation

$\sin^{-1}\sqrt{1-x^2}+\cos^{-1}x=\cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)-\sin^{-1}x$ *is* $[-1,0)$

D) The complete solution set of the equation

$\sin^{-1}\sqrt{1-x^2}+\cos^{-1}x=\cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)-\sin^{-1}x$ *is* $[-1,1]$

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46. Two perpendicular unit vectors $\vec{a}$ and $\vec{b}$ are such that $\left[\vec{r}\ \vec{a}\ \vec{b}\right]=\frac{5}{4}, \vec{r}.\left(3\vec{a}+2\vec{b}\right)=0$ and $\int_{-2\vec{r}.\vec{a}}^{\frac{-4}{3}\vec{r}.\vec{b}} \frac{x+1}{x^2+1}dx=\frac{\pi}{2}$. Which of the following is (are) correct ?

A) $|\vec{r}|^2=\frac{19}{8}$ **B)** $\left|\vec{r}\right|=\frac{7}{4}$

C) $\vec{r}=\frac{\vec{a}}{2}-\frac{3\vec{b}}{4}+\frac{5}{4}(\vec{a}\times\vec{b})$ **D)** $\vec{r}=-\frac{\vec{a}}{2}+\frac{3\vec{b}}{4}+\frac{5}{4}(\vec{a}\times\vec{b})$

47. The sum of real roots of the equation $(2x-1)^2-3|2x-1|+2=0$ is S. then which of the following is/are at least S?

A) 3 **B)** 2 **C)** 1 **D)** 0

48. Let $A_0, A_1, A_2, A_3, A_4, A_5$ are the consecutive vertices of a regular hexagon inscribed in a circle of unit radius. Which of the following holds 'TRUE'.

A) Product of lengths of segments A_0A_1, A_0A_2, A_0A_4 is 3.

B) Area of hexagon $A_0A_1A_2A_3A_4A_5$ is $3\sqrt{3}$

C) Area of triangle $A_0A_1A_2$ *is* $\frac{\sqrt{3}}{4}$

D) Area of quadrilateral $A_0A_2A_3A_5$ *is* $\sqrt{3}$

SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

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49. If $\lim\limits_{x\to 0}\dfrac{\left(1-\cos\left(1-\cos\dfrac{x}{2}\right)\right)}{2^m x^n}$ is equal to the lefthand derivative of $e^{-|x|}$ at x = '0'. Then find the value of $\left(\dfrac{n-10m}{10}\right)$.

50. If z_1, z_2 are complex numbers, then the maximum value of $\dfrac{z_1\overline{z_2}+\overline{z_1}z_2+z_1z_2+\overline{z_1}\,\overline{z_2}}{|z_1z_2|}$ is equal to ____

51. Let $f(x)=ax^2+bx+c.$ where $a\neq 0,\ a,b,c$ are integers and f(1) = 1, $6 < f(3) < 8$ and $18 < f(5) < 22$. Then the number of solutions of equation $f(x) = e^x$ is ___

52. Let $L=\lim\limits_{x\to\infty}\left(\left(\dfrac{x+1}{x-1}\right)^x-e^2\right)x^2$, then the value of $\dfrac{9L}{2e^2}$ is equal to ____

53. Let f:$[0,100]\to[0,100]$ $f(x)=[x]+\{x\}^2$. *Let* $g(x)=f^{-1}(x)$, then sum of all solutions of $g'(x)=1$ in $x\in[0,100]$ is___ ([.] is G.I.F and {} is F.P.F)

54. Let $\left(2+\dfrac{x^4}{x^2+1}\right)^{10}=A_{20}x^{20}+A_{19}x^{19}+A_{18}x^{18}+.....A_1x+A_0+\dfrac{B_1}{x^2+1}+\dfrac{B_2}{\left(x^2+1\right)^2}+.....\dfrac{B_{10}}{\left(x^2+1\right)^{10}}$, $A_{20}, A_{19}, A_{18}, ...A_1, A_0, B_1, B_2, ...B_{10}$ are constants then find A_0.

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31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

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ALL INDIA OPEN CATEGORY RANKS



BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

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